Two-Track Closure for the Three-Petal Sunflower Endpoint

Generated Blocker-Incidence Paths and Kernel-Governed Seed Capacity

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Abstract

Let $\mathcal{F}$ be a finite n-uniform set family. This paper proves the three-petal sunflower endpoint by a sequential two-track argument. The combinatorial track constructs a cardinality-minimal blocker with private edges and a sunflower-free residual tower. Its source-by-source continuation is explicit: for $x_j \in B^{(j)}$, the next-coordinate set is

$$B^{(j+1)} \cap R^{(j)}_{x_j}.$$

This set is nonempty because $R^{(j)}_{x_j}$ is an edge of the next residual family and $B^{(j+1)}$ hits every such edge. Iteration gives a finite branching path relation from every initial blocker coordinate to at least one literal coordinate of the controlled terminal blocker. No Hall matching, canonical successor, or identification of a residual edge with a blocker point is used.

In the controlled seed range every minimal blocker has cardinality at most

$$(3-1)\cdot 2047 = 4094$$

and therefore admits an actual family-local embedding into $\mathrm{Fin}(4094)$. The number 4094 is generated entirely by this combinatorial seed theorem.

The AASC track begins only after this relation is populated. It takes the standing-reach relation as a subrelation of generated reachability and proves four endpoint facts in each fixed support/profile/role cell: at least one generated reach remains standing; two standing terminal carriers for one source are equal; a shared standing carrier determines one complete endpoint type; and same complete type is skin, so private-witness finality identifies the sources. Unique existence therefore defines an injective map from the initial cell to the literal terminal blocker. Composing with the ordinary seed embedding gives an injection into $\mathrm{Fin}(4094)$.

Thus each fixed support/profile/role fibre has at most 4094 members. There are 2^6 observational profiles and four blocker roles in each of the eight ambient support cells, so the effective blocker has size at most

$$8 \cdot 2^6 \cdot 4 \cdot 4094 = 8{,}384{,}512.$$

The ordinary one-point-link recurrence then gives

$$|\mathcal{F}| \le 8{,}384{,}512^n.$$

The theorem is not relative to an elective "AASC regime." No AASC-side predicate restricts the family domain. Determinate family identity, comparison, admissible transformation, standing-bearing endpoint use, and act-time finality instantiate Admissibility, Standing, Reference, and Irreversibility by the kernel necessity theorem. Combinatorics generates all positive finite candidates; AASC performs exhaustion and impossibility on their endpoint use; the final numeral is only a readout of a literal carrier.

The public Lean companion now verifies the generated next-coordinate relation, its source-by-source nonemptiness, the dependent path tower, literal terminal carrier projection, preservation

of the no-sunflower countercase, the relation-level kernel bridge interface, choice independence, terminal-carrier injectivity from that interface, and the resulting 4094 bound. The sunflower-specific proof that instantiates the four governance fields remains the explicit manuscript-to-Lean boundary; it is not hidden as an axiom or misreported as an AASC-free combinatorial theorem.

Mathematical status and reading lock

No framework-relative qualification. The theorem quantifies over every finite uniform family. The kernel is not an optional predicate imposed on a pre-existing subdomain: determinacy, identity, comparison, licensed transformation, standing-bearing endpoint use, and irreversible report finality are the neutral necessities of the theorem act itself.

Positive generation is combinatorial. For every current blocker coordinate x_j, Track I defines the complete finite successor set $B^{(j+1)} \cap R^{(j)}_{x_j}$ and proves it nonempty. Iteration constructs all source-labelled paths and all terminal carriers before AASC is invoked.

No residual–coordinate conflation. A residual $R^{(j)}_{x_j}$ is an edge of the next family, not a point of the next blocker. A successor is a point x_{j+1} in the literal intersection $R^{(j)}_{x_j} \cap B^{(j+1)}$. Branching is retained as a relation; no Hall matching or canonical representative is assumed.

AASC is eliminative. AASC does not create a path, seed carrier, or finite code. On the already populated reachability relation it proves that a standing reach survives, that one source has one standing terminal carrier, and that one carrier cannot host two distinct same-cell standing source identities.

The collision theorem is the crossing. Within a fixed support/profile/role cell, a shared standing terminal carrier fixes the complete endpoint type. Any remaining standing-bearing difference is an excluded split, target change, or independent authority; otherwise it is skin. Private-witness finality turns source skin-equivalence into literal equality.

Combinatorial origin of 4094. The cutoff 2047, the estimate $2 \cdot 2047 = 4094$, and the embedding of the controlled terminal blocker into $\mathrm{Fin}(4094)$ are ordinary finite combinatorics. Numerical labels are family-local readouts and have no primitive standing.

Lean support boundary. Lean checks the blocker-incidence successor, nonempty dependent path tower, literal terminal projection, no-sunflower transport, relation-level governance interface, carrier injectivity from that interface, and the 4094 consequence. The concrete sunflower-specific proof of the governance interface remains paper-level and is stated as such.

Read this first: proof spine

finite family and fixed sunflower endpoint

$\Downarrow$

minimal blocker, private edges, residual family

$\Downarrow$

$$\boxed{x_j \mapsto B^{(j+1)} \cap R^{(j)}_{x_j} \neq \varnothing}$$

$\Downarrow$

all generated paths $\mathsf{Path}_m(x)$ and terminal relation Reach_U

$\Downarrow$ kernel exhaustion on generated pairs

left-total, single-valued standing subrelation

$\Downarrow$ same-carrier collision impossibility

$c_U : U \hookrightarrow B_{\min}(\mathcal{F}^{(m)})$

$\Downarrow$ ordinary seed embedding

$\mathrm{Fin}(4094)$

$\Downarrow$

support-fibre bound, effective blocker, link recurrence

$\Downarrow$

$|\mathcal{F}| \leq 8{,}384{,}512^n.$

Claim-status ledger

Layer	Sunflower object	Status in the proof
Raw carrier	$X, n, \mathcal{F}, \mathrm{NoSun}_3(\mathcal{F})$	Ordinary finite combinatorial target
Generated witness	$B^{(j)}, E^{(j)}_x, R^{(j)}_x$	Literal finite data
One-step relation	$x \mathsf{Next}_j z \iff z \in R^{(j)}_x \cap B^{(j+1)}$	Combinatorially generated and left-total
Path relation	$\mathsf{Path}_m(x)$ and $\mathsf{Reach}_U(x, z)$	All branches retained; no selector
Standing subrelation	$\mathsf{StReach}_U \subseteq \mathsf{Reach}_U$	AASC classification of generated pairs
Standing carrier	$c_U(x) = \iota z\, \mathsf{StReach}_U(x, z)$	Defined after existence and uniqueness
Collision closure	$c_U(x) = c_U(y) \Rightarrow x = y$ in one cell	Kernel exhaustion plus private-witness finality
Controlled seed	$B_{\min}(\mathcal{F}^{(m)})$ at rank at most 2047	Literal combinatorial carrier
Seed readout	$B_{\min}(\mathcal{F}^{(m)}) \hookrightarrow \mathrm{Fin}(4094)$	Family-local finite embedding
Fibre inequality	$\lvert \mathrm{supp}^{-1}(S) \rvert \leq 2^6 \cdot 4 \cdot 4094$	Counting after generated carrier injectivity
Lean object	`KernelFaithfulStandingPathBridge`	Explicit remaining internalization boundary

Contents

1. Claim, determinate theorem domain, and dependency order

1.1. The endpoint theorem

Definition 1.1 (Uniform family and three-sunflower)**.** *Let X be a finite ground set. A family $\mathcal{F} \subseteq \binom{X}{n}$ is n-uniform. Three distinct members $A_0, A_1, A_2 \in \mathcal{F}$ form a* three-sunflower *if there is a core C such that*

$$A_i \cap A_j = C \qquad (i \neq j).$$

Equivalently, the petals $A_i \setminus C$ are pairwise disjoint.

Theorem 1.2 (Three-petal endpoint)**.** *For every finite ground set X, every $n \in \mathbb{N}$, and every n-uniform family $\mathcal{F} \subseteq \binom{X}{n}$,*

$$\mathrm{NoSun}_3(\mathcal{F}) \quad \Longrightarrow \quad |\mathcal{F}| \leq 8{,}384{,}512^n.$$

Consequently, $|\mathcal{F}| > 8{,}384{,}512^n$ forces a three-sunflower.

The constant is not offered as an asymptotic improvement over the strongest modern sunflower bounds [1, 14, 2]. The contribution is the closure architecture: literal private-witness generation, strict rank descent, a generated blocker-incidence path relation, endpoint collision exhaustion on that relation, family-local finite readout, and the ordinary link recurrence.

1.2. Determinate theorem domain and context non-smuggling

Let $D = D(\mathcal{F}, n)$ denote the endpoint package consisting of the finite ground set, submitted family, predicate NoSun_3, maximal and minimal blockers, private witnesses, residual families, generated incidence paths, support/profile/role comparison cells, standing-reach classification, and the final endpoint report.

Proposition 1.3 (Sunflower context non-smuggling)**.** *The package D fixes the carrier and the generated finite operations under evaluation. It does not assert:*

single-valuedness of the standing subrelation,
injectivity of its terminal carrier map,
$|\operatorname{supp}^{-1}(S)| \leq 1{,}048{,}064,$
$|\mathcal{F}| \leq 8{,}384{,}512^n.$

and it does not assert the existence of a sunflower.

Proof. The blockers, private edges, residual identities, local residual slots, rank tower, successor intersections, and complete path relation are generated data. Standing-subrelation population, choice independence, collision exclusion, the finite slot, and the endpoint are derived in Sections 5–7. □

Scope Statement 1.4 (Alternative proof classes). The paper does not claim that removing the kernel crossing leaves a conventional combinatorial proof. A conventional theorem yielding the same high-rank capacity inheritance would be an alternative proof or translation, not a missing family-side hypothesis of the present theorem.

1.3. The exact crossing

The tracks are sequential:

$$
\begin{array}{c}
\text{finite family and fixed endpoint} \\
\Downarrow \\
\text{core–petal geometry, matching, blocker, private witnesses} \\
\Downarrow \\
\text{residual tower and populated blocker-incidence path relation} \\
\Downarrow \\
\boxed{\text{standing-subrelation population and collision exclusion}} \\
\Downarrow \\
\text{generated carrier injection into the terminal blocker} \\
\Downarrow \quad \text{ordinary seed readout} \\
\mathrm{Fin}(4094) \\
\Downarrow \\
\text{support-fibre exclusion, effective blocker, recurrence} \\
\Downarrow \\
|\mathcal{F}| \le 8{,}384{,}512^n.
\end{array}
$$

Everything through population of the terminal reachability relation and the seed blocker bound is finite set theory. The kernel neither creates a path nor assigns semantic content to a numerical slot. Its load-bearing work is to prove standing-subset survival, choice independence, and impossibility of a same-carrier collision inside a fixed complete comparison cell.

1.4. Explicit, source-neutral kernel import

Definition 1.5 (Kernel-neutral sunflower endpoint adequacy)**.** *The package D is target-adequate when the following hold before theorem- bearing use.*

(A1) ***Target determinacy.*** *Ground set, rank, family, sunflower predicate, private-witness identity, comparison class, and endpoint are fixed.*

(A2) ***Step-status evaluability.*** *Matching, blocker reduction, residual formation, quotienting, comparison, and reconstruction each have a determinate status as licensed continuation, support, obstruction, failure, or scope change. No finite capacity or injection is asserted here.*

(A3) ***Act-time finality.*** *Later recoding, selector data, or bridge data creates a new certified act; it does not repair earlier standing or reference loss as the same act.*

(A4) ***Same-regime fidelity.*** *Lawful redescription preserves the family, private-parent relation, standing-bearing provenance, endpoint role, and report status. Failure is package change rather than continuation of the same countercase.*

Write

$$\mathsf{TargetAdeq}(D) = \mathsf{TargetDet}(D) \wedge \mathsf{StepEval}(D) \wedge \mathsf{ActFinal}(D) \wedge \mathsf{SameReg}(D).$$

A package is $\mathsf{NonDeg}(D)$ *when it is target-adequate, contains a genuine standing-bearing endpoint act, and is not a purely syntactic trace in which all distinctions collapse.*

Definition 1.6 (Sunflower kernel)**.**

$$\mathcal{K}_{\mathrm{sun}}(D) = \mathsf{Adm}_D \wedge \mathsf{St}_D \wedge \mathsf{Ref}_D \wedge \mathsf{Irr}_D.$$

Theorem 1.7 (Kernel necessity for the sunflower endpoint)**.** *For every fixed sunflower endpoint package D,*

$$\mathsf{NonDeg}(D) \Longrightarrow \mathcal{K}_{\text{sun}}(D).$$

Import and local replay. The fixed-domain theorem for non-degenerate construction is established in [11]. Target determinacy supplies Reference; step-status evaluability supplies standing-bearing classification; act-time finality supplies Irreversibility; and same-regime fidelity supplies the Admissibility boundary between lawful continuation and package change. The AASC vocabulary names the consequence theory of these forced roles; it does not add a family- side condition. □

Corollary 1.8 (Kernel concession)**.** *Any assessment retaining (A1)–(A4) and theorem-bearing endpoint use has already retained the functional content of $\mathcal{K}_{\text{sun}}(D)$, whether or not it uses AASC terminology.*

Theorem 1.9 (Framework-relativity exclusion)**.** *Suppose a purported non-AASC version preserves the family, private-witness identity, standing-bearing continuation, comparison class, and endpoint. Then it either instantiates $\mathcal{K}_{\text{sun}}(D)$, reproduces the same work under another name, loses the target or endpoint act, or changes the domain, carrier, evaluator, or policy. There is no additional AASC-regime predicate restricting Theorem 1.2.*

Proof. If the neutral antecedent and endpoint use are preserved, Theorem 1.7 applies. A differently named package that performs the same work is governance-equivalent. Otherwise an item of (A1)–(A4), the endpoint act, or the standing-bearing identity bundle has changed. □

Corollary 1.10 (No qualified framework acceptance)**.** *The verdict "valid only within AASC" is unavailable while the ordinary determinate theorem package is retained. A correctness objection must attack a named kernel theorem, generated-path population, standing-subrelation population, single-valuedness, same-carrier collision exclusion, the finite counting, or the recurrence.*

1.5. Positive generation versus eliminative closure

The proof observes a strict dependency boundary.

Combinatorial generation	AASC exhaustion and impossibility
minimal blocker and private edges	standing/reference status of generated paths
successor intersections $R_x^{(j)} \cap B^{(j+1)}$	no contribution; this is positive generation
all finite source-labelled paths	survival of a standing generated reach
literal terminal carrier projection	choice independence and collision impossibility
seed embedding into Fin(4094)	no promotion of readout labels to primitive carriers

The boundary-trace normal form is used only to identify the admissible source of role asymmetry. The private-edge lineage and residual-fibre position are already declared internal/provenance structure, so no environmental boundary trace or canonical representative is imported [6, 5]. The residue- before-readout discipline is the same one made explicit in the canonical role-projection analysis of [7]: typing and standing precede coordinate display.

1.6. Finite certificate and proof responsibility

Definition 1.11 (Constraint-saturated seed-capacity certificate)**.** *For a fixed support/profile/role fibre U, the certificate $\mathfrak{C}_{\mathrm{sun}}(D, U)$ contains:*

$$(U,\ \mathsf{Next}_j,\ \mathsf{Path}_m,\ \mathsf{Reach}_U,\ \mathsf{StReach}_U,\ c_U,\ G_{\mathrm{seed}}).$$

The first four fields are finite combinatorial data. The standing relation is a subrelation of generated reachability; c_U is constructed only after its existence and uniqueness are proved; and $G_{\mathrm{seed}} = \mathcal{F}^{(m)}$. The certificate contains no fibre bound, finite global slot, bounded-load object, or endpoint estimate.

Table 2: Certificate provenance and proof responsibility.

Certificate field	Generated by	Closed by	Exact use
Private-witness identity	minimal blocker and private edge	residual-fibre injection	literal source distinction
One-step successor relation	next residual edge and next blocker	blocker hits every edge	nonempty finite successor set
Terminal path relation	iteration of successor relation	finite induction	all generated terminal candidates
Standing subrelation	generated reachability	no-standing-sink on populated locus	at least one live carrier per source
Single-valuedness	two standing terminal carriers	split/target/authorizer exhaustion	choice-independent carrier
Carrier injectivity	shared carrier in one fixed cell	complete-type and skin finality	preserves source capacity
Seed numerical readout	seed blocker estimate	finite injection	4094 family-local labels

The number 4094 appears only in the last row. The prior rows construct the candidate population and prove that the standing carrier assignment neither depends on a path choice nor merges two same-cell sources.

Audit Gate 1.12 (Dependency-order test)**.** *Every kernel invocation must name a pair already in Reach_U, its fixed same-scope use, and the exact excluded disposition. Every finite-cardinality claim must name a literal finite carrier or injection. A numerical slot may be introduced only after the generated terminal carrier map is proved injective.*

2. Track I: the finite combinatorial carrier

2.1. Core–petal normal form

For $C \subseteq X$ define the link over C by

$$\mathcal{F}_C = \{A \setminus C : A \in \mathcal{F},\ C \subseteq A\}.$$

Write $\nu(\mathcal{G})$ for the maximum number of pairwise disjoint members of a finite family $\mathcal{G}$.

Proposition 2.1 (Core–petal equivalence)**.** *For every $k \geq 2$,*

$$\mathrm{Sun}_k(\mathcal{F}) \iff \exists C \subseteq X\ (\nu(\mathcal{F}_C) \geq k).$$

In particular,

$$\mathrm{NoSun}_k(\mathcal{F}) \iff \forall C \subseteq X\ (\nu(\mathcal{F}_C) < k).$$

Proof. If $A_0, \dots, A_{k-1}$ have common pairwise intersection C, then $A_i \setminus C$ are pairwise disjoint members of $\mathcal{F}_C$. Conversely, pairwise disjoint $P_i \in \mathcal{F}_C$ lift to $C \cup P_i \in \mathcal{F}$, and the lifted sets have pairwise intersection exactly C. □

This equivalence fixes the endpoint carrier. A sunflower is not inferred from a semantic label: it is a concrete matching of petals in a concrete link.

2.2. Maximal matching and the raw blocker

Fix a core C and a maximal pairwise disjoint subfamily $\mathcal{M} \subseteq \mathcal{F}_C$. Put

$$B_{\mathrm{raw}} = \bigcup_{P \in \mathcal{M}} P.$$

Lemma 2.2 (Maximal-matching blocker). *Every nonempty $P \in \mathcal{F}_C$ meets B_{raw}. If $\mathrm{NoSun}_k(\mathcal{F})$ and each link petal has size at most r, then*

$$|\mathcal{M}| \le k-1, \qquad |B_{\mathrm{raw}}| \le (k-1)r.$$

Proof. A petal disjoint from B_{raw} would extend $\mathcal{M}$, contrary to maximality. By Proposition 2.1, $\mathcal{M}$ has fewer than k members. The union bound gives the cardinal estimate. □

At the empty core and rank n, this is the standard rank-dependent Erdős–Rado hitting set. It is a literal blocker, but it does not provide an effective constant-size blocker because its bound grows with n.

2.3. Why the proof is density-restricted

No fixed constant can bound a literal hitting blocker for every three-sunflower-free family at every rank. On the ground set $V \times V$, let

$$S_i = (\{i\} \times V) \cup (V \times \{i\}) \qquad (i \in V).$$

If $i \neq j$, then

$$S_i \cap S_j = \{(i,j),(j,i)\}.$$

For three distinct indices these pairwise intersections are different, so the family $\{S_i : i \in V\}$ contains no three-sunflower. Any hitting set $D \subseteq V \times V$ for all pair-stars must nevertheless cover every vertex of V as a first or second coordinate. Hence

$$|V| \le 2|D|.$$

Taking $|V|$ arbitrarily large defeats every proposed fixed all-family literal blocker bound. This does not refute the sunflower lemma; it refutes the stronger and unnecessary assertion that a fixed literal blocker is available before the quantitative countercase is imposed.

Accordingly, all later population sources are requested only in a *dense countercase*:

$$|\mathcal{F}| > B^n.$$

If this inequality fails, the desired endpoint estimate already holds. This restriction is exact and prevents a false all-family source from entering the proof.

Audit Gate 2.3 (Star-family regression). *A step asserting that every sunflower-free family has a literal hitting blocker of size at most B is invalid. The permitted conclusion is that every dense endpoint countercase has an effective blocker of size at most B, and that conclusion is obtained only after the support-fibre closure.*

2.4. Canonical support cells and the conservative ambient alphabet

Fix the empty-core maximal matching used to form the canonical raw blocker. Its petals are pairwise disjoint. Consequently, every coordinate of B_{raw} belongs to exactly one matching petal. After embedding the at most $k-1$ petals into $\{0,\ldots,k-1\}$, encode that unique incidence as a singleton subset. The formal carrier therefore uses

$$\mathrm{supp} : B_{\min} \longrightarrow \mathcal{P}(\{0,1,2\})$$

for $k=3$.

Lemma 2.4 (Singleton support-cell realization)**.** *For every $x \in B_{\min}$, the set* $\mathrm{supp}(x)$ *has cardinality one. Thus only singleton cells are realized, although the formal codomain is the full eight-element powerset $\mathcal{P}(\{0,1,2\})$.*

Proof. The minimal blocker is contained in the raw blocker. A raw-blocker coordinate lies in at least one matching petal by construction and in at most one because the matching petals are pairwise disjoint. Its encoded support is therefore a singleton. □

The full powerset is retained as the uniform ambient support alphabet used by the Lean interface. Counting all eight ambient cells is conservative: fibres over unrealized cells are empty. The local theorem bounds each fibre $\mathrm{supp}^{-1}(S)$, and the global blocker bound sums over the complete ambient alphabet. No quotient representative replaces the literal blocker, and no live edge hit is discarded.

3. Minimal blockers, private witnesses, and generated incidence paths

3.1. Cardinality-minimal blocker

Let B_{raw} be the finite blocker from the empty-core maximal matching. Choose a hitting subset $B \subseteq B_{\mathrm{raw}}$ of minimum cardinality and write $B_{\min} = B$. The choice is made inside a finite nonempty collection; no representative-selection rule receives standing-bearing force.

Lemma 3.1 (Private edge)**.** *For every $x \in B$ there is an edge $E_x \in \mathcal{F}$ satisfying*

$$E_x \cap B = \{x\}.$$

Moreover, $x \mapsto E_x$ is injective.

Proof. If no such edge existed, $B \setminus \{x\}$ would still hit every edge, contradicting cardinality minimality. If $E_x = E_y$, then its intersection with B is both $\{x\}$ and $\{y\}$, hence $x = y$. □

Definition 3.2 (Private-witness endpoint load)**.** *For $x \neq y$ in $B_{\min}$, the private-witness endpoint load* $\mathsf{PWLoad}_D(x,y)$ *is the incidence record*

$$E_x, E_y; \qquad x \in E_x, \quad y \notin E_x, \quad y \in E_y, \quad x \notin E_y.$$

Lemma 3.3 (Distinct private witnesses carry a load)**.** *For $x, y \in B_{\min}$,*

$$x \neq y \Longrightarrow \mathsf{PWLoad}_D(x,y).$$

Proof. The positive incidences are part of the private-edge construction. Since $E_x \cap B_{\min} = \{x\}$ and $y \in B_{\min} \setminus \{x\}$, one has $y \notin E_x$; the other negative incidence is symmetric. □

3.2. Skin collapse is literal on the blocker carrier

Define endpoint-incidence equivalence by

$$x \sim_{\text{inc}} y \quad \Longleftrightarrow \quad \forall E \in \mathcal{F}\ (x \in E \Longleftrightarrow y \in E).$$

Proposition 3.4 (Private-witness finality)**.** *For $x, y \in B_{\min}$,*

$$x \sim_{\text{inc}} y \quad \Longleftrightarrow \quad x = y.$$

Consequently, endpoint skin-equivalence on the literal minimal blocker is identity.

Proof. Evaluate incidence on E_x. Since $x \in E_x$ and $E_x \cap B_{\min} = \{x\}$, equivalent incidence forces $y = x$. The reverse implication is immediate. □

3.3. Residual family and local parent slots

For $x \in B_{\min}$ put

$$R_x = E_x \setminus \{x\},$$

and let

$$\mathcal{R}(\mathcal{F}) = \{R_x : x \in B_{\min}\}.$$

This is an $(n-1)$-uniform family. Equal residuals may have several parents, but their multiplicity is bounded.

Lemma 3.5 (Residual-fibre bound)**.** *If $\mathcal{F}$ contains no k-sunflower, then for every residual R,*

$$\big|\{x \in B_{\min} : R_x = R\}\big| < k.$$

For $k = 3$, every residual has at most two parents.

Proof. If distinct $x_0, \dots, x_{k-1}$ had the same residual R, then $E_{x_i} = R \cup \{x_i\}$. The x_i are distinct and absent from R, so the private edges have common pairwise intersection R and form a k-sunflower. □

Fix once and for all a finite enumeration of each residual fibre. Write

$$\operatorname{fibslot}_{R_x}(x) \in \operatorname{Fin}(3)$$

for the resulting local position. This is generated combinatorial data; it is not the later 4094-place seed readout.

Proposition 3.6 (Residual-plus-slot injectivity)**.** *The map*

$$x \longmapsto (R_x, \operatorname{fibslot}_{R_x}(x))$$

is injective on $B_{\min}$.

Proof. Equal first coordinates place the two parents in one residual fibre. Equal local slots then identify the same enumerated parent. □

Definition 3.7 (Native residual distinction)**.** $\mathsf{NativeDist}_D(x, y)$ *means that the difference between x and y is witnessed by private-edge incidence, unequal residuals, unequal local residual slots, a point of $R_x \triangle R_y$, or a declared continuation record.*

Proposition 3.8 (Native content is not independent authority). *A native residual distinction is lawful combinatorial content. By itself it does not create a second target, admissibility gate, primitive carrier source, or endpoint authorizer.*

Proof. The distinction is generated finite data. Endpoint authority is a fixed- domain use role affecting admissibility or theorem status. Raw difference does not supply such authority without a standing-bearing role certificate. □

3.4. Raw private-witness identity

The source object must be typed before any continuation or numerical display.

Definition 3.9 (Raw private-witness identity). *For $x \in B_{\min}$ define*

$$\mathsf{RawId}_D(x) = \big(x, E_x, R_x, \mathrm{fibslot}_{R_x}(x), \mathrm{supp}(x), \mathrm{prof}(x), \mathrm{role}(x)\big).$$

The support, profile, and role are the fixed comparison data of the endpoint package; the private edge, residual, and local slot are generated finite data.

Lemma 3.10 (Raw identity is injective). *The map $x \mapsto \mathsf{RawId}_D(x)$ is injective.*

Proof. It contains the injective residual-plus-slot code of Proposition 3.6; it also retains the private edge and source coordinate. □

Definition 3.11 (Declared private-witness provenance). *The provenance anchor of x is*

$$\mathsf{Prov}_D(x) = \big(E_x, R_x, \mathrm{fibslot}_{R_x}(x)\big).$$

At later ranks it is transported as the finite lineage of private edges, residuals, and local fibre positions generated by the residual operation.

Proposition 3.12 (Provenance is an admitted asymmetry source). *Private-witness provenance is declared before continuation use, stable under lawful identity-preserving redescription, and independent of later success. It therefore occupies the internal/provenance branch of the admissible asymmetry-source normal form. No environmental boundary trace or canonical representative is required to distinguish provenance classes.*

Proof. The private edge and local residual position are determined from the submitted finite family and minimal blocker before any endpoint collision is tested. They are not introduced after failure and are preserved by every same-family redescription that preserves private-parent identity. The role-fixation normal forms of [6, 5] list internal invariants and declared provenance markers before the boundary-trace route. The present provenance occupies those prior branches. □

3.5. Sunflower-free residual tower

Set $\mathcal{F}^{(0)} = \mathcal{F}$. Whenever $\mathcal{F}^{(j)}$ has positive rank, choose a cardinality-minimal blocker $B^{(j)}$ and define

$$\mathcal{F}^{(j+1)} = \mathcal{R}(\mathcal{F}^{(j)}).$$

Write $R_x^{(j)}$ for the residual of $x \in B^{(j)}$.

Lemma 3.13 (Residual sunflower lifting)**.** *If $\mathcal{F}^{(j+1)}$ contains a three-sunflower, then $\mathcal{F}^{(j)}$ contains a three-sunflower. Hence*

$$\mathrm{NoSun}_3(\mathcal{F}^{(0)}) \Longrightarrow \mathrm{NoSun}_3(\mathcal{F}^{(j)})$$

for every stage of the tower.

Proof. Choose the three residual petals and one private parent for each. Reinserting the three distinct private coordinates preserves the common residual core: a private coordinate belongs to no other private residual. The lifted private edges therefore form a three-sunflower. □

At stage j, $\mathcal{F}^{(j)}$ is $(n-j)$-uniform. Thus rank is a strictly decreasing termination measure.

Lemma 3.14 (Strict tower termination)**.** *Let $N = 2047$ and $m = \max(0, n-N)$. The residual tower reaches the common family $\mathcal{F}^{(m)}$ of rank at most N after exactly m steps, and no stage can reopen a higher rank.*

Proof. Residual formation lowers the uniform rank by one. The residual family is constructed from the submitted family and the fixed minimal-blocker operation. Integer rank is therefore a well-founded measure. □

3.6. Generated blocker-incidence continuations

Fix a support cell S, a six-level profile p, and a blocker role r, and put

$$U_{S,p,r} = \{x \in B^{(0)} : \mathrm{supp}(x) = S,\ \mathrm{prof}(x) = p,\ \mathrm{role}(x) = r\}.$$

The residual tower alone is global. To transport one source through it we use the incidence that is already forced by the next literal blocker.

Definition 3.15 (Generated next-coordinate relation)**.** *For $x \in B^{(j)}$ with $j < m$, define*

$$\mathsf{Next}_j(x, z) \quad\Longleftrightarrow\quad z \in B^{(j+1)} \cap R_x^{(j)}.$$

Equivalently, the finite successor set is

$$\mathsf{Next}_j[x] = B^{(j+1)} \cap R_x^{(j)}.$$

The successor is a point of the next blocker lying in the current residual; it is not the residual itself.

Lemma 3.16 (Every generated successor set is inhabited)**.** *For every $j < m$ and every $x \in B^{(j)}$,*

$$\mathsf{Next}_j[x] \neq \varnothing.$$

Proof. By construction $R_x^{(j)}$ is an edge of $\mathcal{F}^{(j+1)}$. The set $B^{(j+1)}$ is a hitting blocker for every edge of $\mathcal{F}^{(j+1)}$. Hence $B^{(j+1)} \cap R_x^{(j)} \neq \varnothing$. □

This one-line incidence is the positive generation previously missing from the high-rank crossing. It uses neither standing, a matching, an injective code, nor a choice of one privileged successor.

Definition 3.17 (Generated incidence path)**.** *For $x_0 \in B^{(0)}$, a generated path of length j is a tuple*

$$P = (x_0, x_1, \ldots, x_j)$$

such that

$$x_i \in B^{(i)}, \qquad x_{i+1} \in R_{x_i}^{(i)} \cap B^{(i+1)} \quad (0 \le i < j).$$

Write $\mathsf{Path}_j(x_0)$ for the finite set of all such paths. All branches are retained as data.

Theorem 3.18 (Source-by-source path population)**.** *For every $x_0 \in B^{(0)}$ and every $0 \le j \le m$,*

$$\mathsf{Path}_j(x_0) \neq \varnothing.$$

In particular, every source reaches at least one literal coordinate of $B^{(m)}$.

Proof. Induct on j. The length-zero path is (x_0). If $(x_0, \dots, x_j)$ has been generated, Lemma 3.16 provides at least one $x_{j+1} \in R^{(j)}_{x_j} \cap B^{(j+1)}$; appending every such coordinate produces the complete next path set. The construction is finite at each stage. It does not assert uniqueness. □

Definition 3.19 (Terminal carrier and generated reachability)**.** *For $P = (x_0, \dots, x_m) \in \mathsf{Path}_m(x_0)$ put*

$$\mathrm{car}(P) = x_m \in B^{(m)}.$$

Define

$$\mathsf{Reach}_U(x, z) \quad \Longleftrightarrow \quad \exists P \in \mathsf{Path}_m(x)\ (\mathrm{car}(P) = z).$$

Thus $\mathsf{Reach}_U \subseteq U \times B^{(m)}$ is a literal finite combinatorial relation and every left fibre is nonempty.

Proposition 3.20 (Generated source injection)**.** *The type of source-labelled generated terminal paths contains an injective copy of $U_{S,p,r}$.*

Proof. Theorem 3.18 supplies at least one path for each source. Pair any chosen path with its source. Equality of the pairs implies equality of their source components. This harmless finite choice displays an already inhabited relation; no selected path is used as the endpoint carrier map. □

Definition 3.21 (Path skin equivalence)**.** *Two generated paths are skin-equivalent when they have the same source and terminal carrier and differ only in route notation, intermediate enumeration, or another feature that changes no fixed reference, standing-bearing role, obstruction, endpoint use, or licensed future use.*

Definition 3.22 (Standing reachability)**.** *For $x \in U$ and $z \in B^{(m)}$, write $\mathsf{StReach}_U(x, z)$ when a path witnessing $\mathsf{Reach}_U(x, z)$ remains standing-bearing in the fixed endpoint act. By definition,*

$$\mathsf{StReach}_U(x, z) \Longrightarrow \mathsf{Reach}_U(x, z).$$

Audit point. Combinatorics has now populated the entire candidate relation before the kernel is invoked. Track II may remove, identify, or exclude generated standing dispositions, but it may not add a pair to Reach_U.

4. The combinatorial seed capacity and the exact high-rank frontier

4.1. Traditional seed population

The classical Erdős–Rado recurrence gives a factorial-type upper bound for fixed k [3]. In the mechanized three-petal specialization, the verified cutoff is

$$N = 2047.$$

Every minimal blocker at rank at most N has cardinality at most

$$(3 - 1)N = 4094.$$

Theorem 4.1 (Seed population)**.** *Let $\mathcal{G}$ be an m-uniform three-sunflower-free family with $m \leq 2047$. There is an injective map*

$$\mathrm{seedCode}_{\mathcal{G}} : B_{\min}(\mathcal{G}) \hookrightarrow \mathrm{Fin}(4094).$$

Proof. The maximal-matching raw blocker has cardinality at most $2m \leq 4094$. A cardinality-minimal blocker is a subset of it and therefore embeds in Fin(4094). This is the traditional seed embedding verified in the Lean companion. □

Remark 4.2 (Capacity, not universal semantics)*.* The map $\mathrm{seedCode}_{\mathcal{G}}$ is family-local. Its values are numerical readouts of the already identified seed carriers. No theorem requires the same numeral in two different seed families to denote one common standing role.

This ordering follows the residue/readout discipline of [7]: an object is first typed in its carrier or role fibre; coordinate projection is then a non-generative display of that object.

4.2. Finite profile and role signature

Let the six endpoint-local constraint levels be architectural foundation, ametric boundary, operator envelope, lawful redescription, composition sequencing, and standing enforcement. For each level ℓ, define $\mathsf{Obs}_\ell(x)$ to be true exactly when the fixed finite endpoint-use record of x contains the corresponding licensed constraint witness. The records are finite and standing is bivalent, so each Obs_ℓ is a total Boolean map. Their joint truth vector gives the concrete profile

$$\mathrm{prof}(x) = \{\ell : \mathsf{Obs}_\ell(x) = 1\} \in \mathcal{P}(\{1, \dots, 6\}),$$

with $2^6 = 64$ possible values. The blocker-role ledger has four values:

$$\mathsf{Skin}, \qquad \mathsf{BoundedFiber}, \qquad \mathsf{TensorSplit}, \qquad \mathsf{SunRole}.$$

The profile/role signature therefore has 256 values.

Here role(x) records whether the endpoint use of x is presentation skin, a bounded carrier fibre, an endpoint-faithful tensor split, or a realized sunflower role. The observations and role are total classification maps. They do not by themselves prove injectivity or finite population. Their load-bearing use is later: within one equal profile/role cell and over one terminal carrier, fixed-domain exhaustion proves equality of complete endpoint type. No such implication is asserted across different cells.

Refining a fixed coarse class by one family-local seed readout gives the within-support capacity

$$Q = 2^6 \cdot 4 \cdot 4094 = 1{,}048{,}064.$$

The three-petal ambient Boolean support alphabet has $2^3 = 8$ cells, so

$$B = 2^3 Q = 8{,}384{,}512.$$

Only singleton support cells are realized by the canonical matching code, but counting all eight ambient cells is conservative and matches the formal interface.

4.3. Reflected classical range

The seeded base dominates the traditional estimate through rank 8,384,511. Thus every three-sunflower-free n-uniform family satisfies

$$|\mathcal{F}| \leq B^n \qquad (n \leq 8{,}384{,}511).$$

The constraint-saturated inheritance crossing is invoked only above this reflected range.

4.4. Finite realization and exact frontier

Let $\mathrm{supp} : B_{\min} \to \mathcal{P}(\{0,1,2\})$ be the canonical support code and write

$$\mathsf{KGF}(\mathcal{F}) \iff \forall S \subseteq \{0,1,2\}, \quad |\,\mathrm{supp}^{-1}(S)| \le Q.$$

Definition 4.3 (Family-local finite realization)**.** *A finite realization is a map*

$$c_D(x) = (\mathrm{prof}(x), \mathrm{role}(x), \mathrm{slot}(x))$$

into

$$\mathcal{P}(\{1,\dots,6\}) \times \{\mathsf{Skin}, \mathsf{BoundedFiber}, \mathsf{TensorSplit}, \mathsf{SunRole}\} \times \mathrm{Fin}(4094)$$

that is injective on each support fibre. The slot component is assembled from family-local seed readouts only after the generated terminal-carrier map has been proved single-valued and injective on each fixed cell.

Theorem 4.4 (Finite realization–fibre equivalence)**.** *A finite realization exists on every canonical support fibre if and only if* $\mathsf{KGF}(\mathcal{F})$ *holds.*

Proof. An injective realization gives the Q-element fibre bound. Conversely, a finite fibre bound gives an embedding into a Q-element set, and the explicit cardinality equivalence with the refined signature transports that embedding. □

Theorem 4.5 (Unconditional finite alternative)**.** *For every three-sunflower-free finite countercase, either a finite local realization exists or there is a canonical support cell S with*

$$Q < |\,\mathrm{supp}^{-1}(S)|.$$

Proof. Apply excluded middle to $\mathsf{KGF}(\mathcal{F})$ and use Theorem 4.4. The ambient support alphabet is finite, so failure produces an explicit cell. □

Sections 5 and 6 prove that every fixed support/profile/role cell injects by a generated terminal-carrier map into the combinatorial seed capacity 4094, thereby excluding the oversized branch.

5. Endpoint governance of the generated incidence relation

5.1. Activation after combinatorial population

Fix a terminal dense high-rank three-sunflower-free package D and one cell $U = U_{S,p,r}$. Track I has already constructed the finite relation

$$\mathsf{Reach}_U \subseteq U \times B^{(m)}$$

and proved that every left fibre is nonempty. Nothing in this section creates a successor, a path, or a terminal blocker coordinate.

Proposition 5.1 (Kernel activation on the generated package)**.** *The package D is non-degenerate and target-adequate. Hence*

$$\mathcal{K}_{\mathrm{sun}}(D) = \mathsf{Adm}_D \wedge \mathsf{St}_D \wedge \mathsf{Ref}_D \wedge \mathsf{Irr}_D$$

is in force on the generated endpoint relation.

Proof. The family, rank, blockers, private edges, path relation, comparison cell, and endpoint are finite and fixed. Generated paths have determinate endpoint status, same-package redescription preserves the private source and terminal carrier, and later repair cannot restore a failed earlier act. Thus the neutral antecedent of Theorem 1.7 holds. ☐

The local consequence packet is used only as follows.

Kernel Principle 5.2 (Fixed reference)**.** *A live path retains its determinate source, terminal carrier, comparison cell, and endpoint act under same-package redescription.*

Kernel Principle 5.3 (Skin has no standing authority)**.** *Route names, intermediate enumerations, and representative choices cannot create, duplicate, or restore standing.*

Kernel Principle 5.4 (Bivalent standing and no standing sink)**.** *Standing is present or absent, not graded. A live standing-bearing source in an instantiated nonempty continuation locus cannot have every generated same-package continuation become standing-null [13, 12, 9].*

Kernel Principle 5.5 (Fixed-domain role exhaustion)**.** *After source, carrier, comparison cell, and endpoint are fixed, any standing-bearing discrepancy is existing anchor/tensor content, an endpoint-faithful split, a target change, or independent authority. Skin supplies no fifth source of standing [8, 10].*

5.2. Disposition is classification, not generation

Definition 5.6 (Generated-path dispositions)**.** *Every already generated path has one of the following endpoint dispositions:*

1. D*: same-package standing reach;*
2. C*: direct combinatorial closeout of the countercase;*
3. T*: endpoint-faithful tensor split;*
4. S*: concrete three-sunflower realization;*
5. X*: target, carrier domain, evaluator, or comparison-cell change;*
6. N*: standing loss, null continuation, or support-only use;*
7. A*: independent authorizer, selector, registry, or same-act repair.*

Theorem 5.7 (Complete generated-path disposition)**.** *Every member of every finite path set* $\mathsf{Path}_m(x)$ *has at least one disposition in Definition 5.6, and no same-scope standing-bearing eighth disposition exists.*

Proof. Bivalence first separates standing use from nonstanding support or failure. For a standing path, anchor/tensor/skin exhaustion separates unchanged same-package use, endpoint-faithful factorization, and presentation. Concrete endpoint realization, target change, and independent authority are recorded separately. These cases give the seven listed dispositions. Fixed-domain exhaustion closes a purported additional same-scope standing source. This classifies the finite candidates constructed in Theorem 3.18; it does not assert that a candidate exists. ☐

5.3. Population of the standing subrelation

Definition 5.8 (Terminal minimal package)**.** *Order countercase packages by uniform rank and then by unresolved endpoint- faithful split complexity. A package is terminal when no smaller-potential direct closeout or admitted tensor split remains.*

Theorem 5.9 (No standing sink on the generated path relation)**.** *For every $x \in U$ there is a terminal blocker coordinate $z \in B^{(m)}$ such that*

$$\mathsf{StReach}_U(x, z).$$

Proof. By Theorem 3.18, the finite candidate locus $\mathsf{Path}_m(x)$ is nonempty. Hence its reachability fibre is nonempty. The literal source x, its private edge E_x, and its use in the fixed no-sunflower endpoint are standing-bearing data, not optional annotations.

If no generated candidate were standing, the complete disposition theorem would place every candidate in direct closeout, split, sunflower, target change, standing loss, or independent repair. Direct closeout and split contradict terminality. Sunflower contradicts NoSun_3, and target change leaves D. Standing loss or repair violates standing conservation, the no-standing-sink theorem, and act-time finality. Therefore at least one already generated reach remains standing. □

The order of this proof is essential: blocker incidence gives the candidate locus before no-standing-sink is applied. The kernel filters a populated finite relation; it does not infer an object from the failure of alternatives.

5.4. Choice independence for one source

Theorem 5.10 (Standing terminal carrier is single-valued)**.** *For $x \in U$ and $z, w \in B^{(m)}$,*

$$\mathsf{StReach}_U(x, z) \wedge \mathsf{StReach}_U(x, w) \quad \Longrightarrow \quad z = w.$$

Proof. Assume $z \neq w$. Since z, w are distinct points of the same literal minimal blocker, terminal private edges separate them by Proposition 3.4; the difference is not skin. Two distinct standing terminal carriers for one fixed source and endpoint therefore require either an endpoint-faithful tensor bifurcation, a target/comparison-cell change, or an independent authorizer selecting two outputs. Those are exactly the T, X, and A branches, excluded in the terminal fixed package. Hence $z = w$. Different generated routes ending at the same z are path skin unless they exhibit one of the same excluded standing-bearing branches. □

By Theorems 5.9 and 5.10, unique existence holds. Define

$$c_U(x) = \text{the unique } z \in B^{(m)} \text{ with } \mathsf{StReach}_U(x, z).$$

This is a genuine function. It is independent of a chosen path representative because single-valuedness is proved at the carrier level.

Theorem 5.11 (The terminal carrier is generated)**.** *For every $x \in U$,*

$$c_U(x) \in B^{(m)} \quad \textit{and} \quad \mathsf{Reach}_U(x, c_U(x)).$$

Proof. The first statement is part of the codomain. The defining standing witness is a generated path witness by Definition 3.22, so the second statement follows. No abstract seed object is assigned to a blocker after the fact: the path already ends at this literal blocker coordinate. □

5.5. Same-carrier collision exhaustion

The finite profile and role are load-bearing here. They are the fixed endpoint-local observations needed to compare two sources that arrive at one carrier. Without equality of those fields, a shared carrier may represent a different tensor or endpoint role and no collision conclusion is drawn.

Theorem 5.12 (Same carrier determines one complete AASC type)**.** *Let $x, y \in U$ and $z \in B^{(m)}$. If*

$$\mathsf{StReach}_U(x, z) \quad \textit{and} \quad \mathsf{StReach}_U(y, z),$$

then the two reaches have the same complete AASC standing type in the fixed endpoint package.

Proof. The terminal carrier, endpoint, support cell, six-level profile, and blocker role agree. Any remaining route or enumeration difference is skin unless it changes standing-bearing endpoint work. If it does change such work, fixed-domain exhaustion places the first difference in extra anchor/tensor content, target change, or independent authority. Extra tensor content is an endpoint-faithful split, target change leaves the comparison package, and independent authority is excluded by fixed-domain finality. Thus no standing-bearing field remains on which the two endpoint types can differ. □

Theorem 5.13 (Kernel-faithful terminal collision exclusion)**.** *The function*

$$c_U : U \longrightarrow B^{(m)}$$

is injective.

Proof. Suppose $c_U(x) = c_U(y) = z$. Both standing reaches exist by definition of c_U. Theorem 5.12 gives equality of their complete AASC types. Same complete type is skin-equivalence: skin has no primitive standing with which to sustain a further distinction. Quotient finality therefore transports the equality back to the fixed source identities. On the literal initial minimal blocker, source skin-equivalence is equality by Proposition 3.4, since E_x contains x and no other blocker coordinate. Hence $x = y$. □

Corollary 5.14 (Terminal carrier capacity)**.** *For every fixed support/profile/role cell U,*

$$|U| \leq |B^{(m)}|.$$

Proof. Apply finite cardinality monotonicity to the injection c_U. □

Theorem 5.15 (Generated standing-carrier embedding)**.** *There is an injection*

$$c_U : U \hookrightarrow B_{\min}(\mathcal{F}^{(m)})$$

whose value at x is the terminal coordinate of an explicitly generated blocker-incidence path from x.

Proof. Generatedness is Theorem 5.11; injectivity is Theorem 5.13. □

> **Audit point.** The old abstract maps ∂_j and ϵ_G are no longer proof inputs. The combinatorial relation is left-total before kernel activation; the kernel proves that its standing subrelation is left-total, single-valued, and collision-free on each fixed cell. The resulting function c_U is then constructed by unique existence.

6. Generated seed-capacity transport and fibre exclusion

6.1. The relation-level bridge

Fix a terminal dense high-rank three-sunflower-free package, one cell

$$U = U_{S,p,r} = \{x \in B^{(0)} : \mathrm{supp}(x) = S,\ \mathrm{prof}(x) = p,\ \mathrm{role}(x) = r\},$$

let $m = \max(0, n - 2047)$, and put $G = \mathcal{F}^{(m)}$.

Theorem 6.1 (Generated standing-path bridge)**.** *The following statements hold simultaneously.*

1. $\mathsf{StReach}_U(x,z)$ *implies* $\mathsf{Reach}_U(x,z)$.
2. *For every* $x \in U$ *there is a* $z \in B_{\min}(G)$ *with* $\mathsf{StReach}_U(x,z)$.
3. *For fixed* x, *two standing terminal carriers are equal.*
4. *Two sources in* U *reaching one standing terminal carrier have the same complete endpoint type, hence are skin-equivalent.*
5. *Source skin-equivalence is literal equality.*

Consequently, unique existence defines an injective generated carrier map

$$c_U : U \hookrightarrow B_{\min}(G).$$

Proof. The first item is definitional. The second is Theorem 5.9; the third is Theorem 5.10; the fourth is Theorem 5.12; and the fifth is Proposition 3.4. The construction and injectivity of c_U are Theorems 5.11 and 5.13. □

This theorem separates the two tracks exactly. The relation and all terminal coordinates in it are generated by blocker incidence. AASC contributes only standing-subset population, choice independence, complete-type determination, and collision impossibility.

6.2. Injection into the controlled seed blocker

Theorem 6.2 (Constraint-saturated seed-capacity transport)**.** *For every fixed support/profile/role cell* U,

$$|U| \le |B_{\min}(G)| \le 4094.$$

More precisely,

$$\mathrm{seedCode}_G \circ c_U : U \hookrightarrow \mathrm{Fin}(4094)$$

is injective.

Proof. The generated carrier map c_U is injective by Theorem 6.1. The family G has rank at most 2047 by Lemma 3.14 and remains three-sunflower-free by Lemma 3.13. Therefore $\mathrm{seedCode}_G$ is the injective combinatorial map of Theorem 4.1. Their composition is injective. □

No conversion of an abstract terminal provenance object into a blocker point occurs. The terminal point is already the last coordinate of the path. The kernel proves that one such point is uniquely standing for each source and cannot be shared by two sources in one complete comparison cell.

6.3. Family-local numerical slots

For the fixed cell U, define

$$\mathrm{slot}_U = \mathrm{seedCode}_G \circ c_U : U \longrightarrow \mathrm{Fin}(4094).$$

Theorem 6.3 (Family-local generated seed slot)**.** *The map* slot_U *is injective. A different enumeration of the same seed blocker changes* slot_U *only by a finite permutation and leaves all equality and cardinality conclusions unchanged.*

Proof. Injectivity is Theorem 6.2. Two enumerations of one finite carrier differ by a permutation. The numeral is downstream readout skin and has no standing authority. □

Apply this construction independently to each finite cell $U_{S,p,r}$. Finite choice combines the cellwise maps into a total function

$$\mathrm{slot} : B_{\min}(\mathcal{F}) \longrightarrow \mathrm{Fin}(4094)$$

whose restriction to each fixed support/profile/role cell is injective. The total slot map need not be injective until the support, profile, and role fields are included.

6.4. Lean-facing private-witness load exhaustion

Theorem 6.4 (Kernel-faithful bounded private-witness population)**.** *For all* $x, y \in B_{\min}(\mathcal{F})$*, equality of support, profile, and role together with the private-witness load implies*

$$\begin{aligned} \mathrm{slot}(x) \neq \mathrm{slot}(y) \quad & \vee \quad \mathsf{PrimeSplit}_D(x, y) \\ & \vee \quad \mathrm{Sun}_3(\mathcal{F}) \\ & \vee \quad \exists a \ \mathsf{IndAuth}_D(a). \end{aligned}$$

In the terminal fixed three-sunflower-free package, the first branch holds.

Proof. The equal coarse fields place x and y in one cell U. The private-witness load gives $x \neq y$. The map slot_U is injective by Theorem 6.3, so $\mathrm{slot}(x) \neq \mathrm{slot}(y)$. The stated disjunction follows. □

The theorem constructs the existing Lean bounded-load interface downstream of the generated standing-path bridge. It does not supply continuation population to that bridge.

6.5. Finite-code collision and injectivity

Define

$$c_D(x) = (\mathrm{prof}(x), \mathrm{role}(x), \mathrm{slot}(x)).$$

Theorem 6.5 (Same-support finite-code injectivity)**.** *For* $x, y \in B_{\min}(\mathcal{F})$ *in a terminal dense high-rank three-sunflower-free countercase,*

$$\mathrm{supp}(x) = \mathrm{supp}(y) \quad \textit{and} \quad c_D(x) = c_D(y) \quad \Longrightarrow \quad x = y.$$

Proof. Equal codes place x and y in one cell and give equal values of its injective slot map. Hence $x = y$. □

The implication order is deliberate. The finite code is assembled only after the terminal carrier relation has been populated, governed, and proved collision-free.

6.6. The exact fibre theorem

Theorem 6.6 (Kernel-Governed Same-Support Fibre Exclusion)**.** *For every canonical support cell S in a terminal dense high-rank three-sunflower-free countercase,*

$$|\operatorname{supp}^{-1}(S)| \le 1{,}048{,}064.$$

Proof. Partition $\operatorname{supp}^{-1}(S)$ by its 2^6 six-level profiles and four roles. Each cell has cardinality at most 4094 by Theorem 6.2. Therefore

$$|\operatorname{supp}^{-1}(S)| \le 2^6 \cdot 4 \cdot 4094 = 1{,}048{,}064.$$

Equivalently, Theorem 6.5 embeds the support fibre into the refined finite signature of that cardinality. □

The factors 2^6 and 4 are not decorative: equality of profile and role is used precisely in Theorem 5.12 to obtain complete same-carrier endpoint-type equality. No such conclusion is asserted across different cells.

6.7. Locality of the crossing conclusion

The crossing proves exactly

$$\forall S \subseteq \{0,1,2\}, \qquad |\operatorname{supp}^{-1}(S)| \le 1{,}048{,}064.$$

Section 7 uses this inequality to obtain a literal effective blocker and then applies the ordinary one-point-link recurrence.

7. Assembly of the endpoint pipeline

7.1. From local fibres to an effective blocker

The ambient Boolean support alphabet has $2^3 = 8$ cells. Applying Theorem 6.6 to each cell gives

$$|B_{\min}| = \sum_{S\subseteq\{0,1,2\}} |\operatorname{supp}^{-1}(S)| \le 8 \cdot 1{,}048{,}064 = 8{,}384{,}512 = B.$$

Fibres over unrealized support cells are empty by Lemma 2.4. Because $B_{\min}$ is a literal hitting blocker, no quotient reconstruction is required.

Proposition 7.1 (Dense effective blocker)**.** *Every terminal dense three-sunflower-free countercase has a hitting blocker of cardinality at most $B = 8{,}384{,}512$.*

Proof. Use the cardinality-minimal blocker constructed from the empty-core maximal matching and the support-cell sum above. □

If a nonterminal package is encountered, perform its admitted endpoint- faithful tensor factorization first. The factors have smaller reduction potential and their endpoint estimates reconstruct the estimate for the package. Every minimal countercase is therefore terminal.

7.2. One-point-link recurrence

Let $f(n)$ be the maximum size of a finite three-sunflower-free n-uniform family, with $f(0) = 1$. In a dense countercase choose the effective blocker D. For $x \in D$, define

$$\mathcal{F}(x) = \{E \setminus \{x\} : E \in \mathcal{F},\ x \in E\}.$$

Each $\mathcal{F}(x)$ is $(n-1)$-uniform and three-sunflower-free: a sunflower after deletion lifts by reinserting the common point x. Since every edge meets D,

$$|\mathcal{F}| \le \sum_{x \in D} |\mathcal{F}(x)| \le |D| f(n-1) \le B f(n-1).$$

Repeated induction gives $f(n) \le B^n$.

7.3. Proof of the main theorem

Proof of Theorem 1.2. Proceed by strong induction on n. For $n \le 8{,}384{,}511$, the reflected classical estimate in Section 4 gives the result. Let n be larger, assume the result at every lower rank, and suppose for contradiction that $\mathrm{NoSun}_3(\mathcal{F})$ and $|\mathcal{F}| > B^n$.

Choose a countercase minimal under the reduction potential and exhaust every endpoint-faithful tensor split. The resulting package is terminal. Track I constructs the minimal private-witness blocker, the residual-plus-local-slot identities, the sunflower-free residual tower, and the source-by-source blocker-incidence path relation. Every source reaches at least one literal coordinate of the controlled seed blocker before the kernel is invoked.

Proposition 5.1 instantiates the fixed-domain kernel. Theorem 5.9 proves that the standing subrelation is left-total, Theorem 5.10 proves it single-valued, and Theorem 5.13 excludes a shared standing carrier inside one fixed support/profile/role cell. Unique existence therefore constructs the generated injection c_U into the literal terminal blocker. The ordinary seed embedding then gives Theorem 6.2.

Theorem 6.4 constructs the Lean-facing bounded-load object, and Theorem 6.6 excludes the oversized support fibre. Proposition 7.1 supplies a literal blocker D with $|D| \le B$. The one-point links have lower rank, so the inductive hypothesis gives

$$|\mathcal{F}| \le |D| B^{n-1} \le B^n,$$

contradicting size excess. Hence no countercase exists. □

7.4. Where each component stops

The finite combinatorial component generates the family, private loads, residual identities, local residual slots, rank tower, every blocker-incidence path, every terminal coordinate occurring in such a path, the controlled seed family, and the seed embedding. It does not by itself determine which generated terminal reaches remain standing-bearing or exclude a collision between two same-cell sources at one terminal carrier.

The AASC consequence component filters the generated relation, proves standing-subset population, choice independence, same-carrier complete-type equality, and collision impossibility. It does not generate a path, a terminal carrier, or 4094, and it does not promote numerical labels to standing. This component stops at the support-fibre inequality. The literal blocker, one-point-link recurrence, and concrete sunflower endpoint remain ordinary finite combinatorics.

Corollary 7.2 (Existence form)**.** *If $|\mathcal{F}| > 8{,}384{,}512^n$, then there are distinct $A_0, A_1, A_2 \in \mathcal{F}$ and a core C with*

$$A_0 \cap A_1 = A_0 \cap A_2 = A_1 \cap A_2 = C.$$

The witness is the ordinary combinatorial consequence of eliminating every no-sunflower countercase. It is not selected by the kernel consequence layer.

8. Lean correspondence and formalization boundary

8.1. What the public development checks

The companion repository [4] now contains both the finite private-witness reduction and the generated blocker-incidence tower.

Manuscript object	Lean declaration or checked surface
Residual R_x	`PrivateWitnessReduction.residual`
Residual family	`PrivateWitnessReduction.residualFamily`
Residual sunflower lifting	`residualFamily_noSunflower`
Residual-fibre bound	`residualFiber_card_lt`
Residual-plus-slot injection	`residualSlotCode_injective`
Generated successor type	`GeneratedIncidenceTower.NextCoordinate`
Successor population	`GeneratedIncidenceTower.nextCoordinate_nonempty`
Generated one-step relation	`GeneratedIncidenceTower.NextRel`
Dependent path tower	`GeneratedIncidenceTower.IncidencePath`
Source-by-source path population	`GeneratedIncidenceTower.incidencePath_nonempty`
Literal terminal carrier	`IncidencePath.terminalCoordinate`
Generated reachability	`GeneratedIncidenceTower.ReachesTerminalCarrier`
Reachability population	`exists_reachesTerminalCarrier`
Terminal no-sunflower transport	`terminalFamily_noSunflower`
Relation-level kernel bridge	`KernelFaithfulStandingPathBridge`
Standing carrier is generated	`standingCarrier_reachable`
Choice independence	`standingCarrier_choice_independent`
Terminal-carrier injectivity	`standingCarrierOf_injective`
Generated seed code	`KernelFaithfulStandingPathBridge.seedCode`
Cell bound 4094	`KernelFaithfulStandingPathBridge.cell_card_le_4094`
Endpoint transfer	`sunflower_of_seeded_` `rankDeletionPopulationInheritance`

The crucial new theorem is unconditional combinatorial population:

> for every initial minimal-blocker coordinate, the dependent blocker-incidence path type to the cutoff rank is nonempty.

The proof uses only that the next minimal blocker hits the current residual edge. Its result is a relation, not an injective map, and all branching is retained.

8.2. The relation-level kernel interface

The Lean structure `KernelFaithfulStandingPathBridge` receives no path-population field. That theorem has already been proved by combinatorics. Its fields state only:

1. a standing relation on source and terminal blocker coordinates;

2. proof that every standing pair belongs to generated reachability;
3. nonemptiness of the standing fibre for each source;
4. single-valuedness of each standing fibre;
5. same-carrier determination of one AASC quotient type; and
6. quotient finality back to literal source equality.

From those fields Lean constructs the unique standing carrier, proves it is reachable by a generated path, proves independence from any standing witness, proves injectivity, composes with the traditional terminal-blocker embedding, and obtains the literal bound 4094.

This placement matters. AASC is not asked to construct `IncidencePath`; combinatorics is not credited with proving endpoint collision impossibility.

8.3. What remains paper-level

The four sunflower-specific governance conclusions in Section 5 are proved in the manuscript but are represented in Lean by the fields of `KernelFaithfulStandingPathBridge`. The current build does not yet construct an unparameterized instance of that structure from the imported kernel corpus.

Accordingly, compilation verifies:

- the complete positive generation demanded by the continuation objection;
- the exact typed boundary between generated reachability and standing reachability;
- every construction and numerical consequence after the governance fields; and
- that no hidden Lean axiom, placeholder, support-fibre bound, or endpoint bound is used in the checked path construction.

Compilation does not by itself certify the prose proof of standing-fibre population, single-valuedness, or same-carrier complete-type exhaustion. Those are the next internalization target, not silently accepted family hypotheses.

8.4. Manuscript-faithful internalization target

The decisive future declaration has the following shape:

```
noncomputable def standingPathBridge_of_generatedData
    (F : UniformSetFamily alpha (baseRank + steps + 1))
    (hSun : Not (HasSunflower 3 F))
    (cell : Finset (InitialCarrier F))
    (hTerminal : TerminalCountercase F cell)
    (hKernel : KernelFaithfulEndpointUse F cell) :
    KernelFaithfulStandingPathBridge F cell := by
  -- instantiate only the four endpoint-governance obligations
  ...
```

Its assumptions must not include, directly or equivalently, the support-fibre bound, an injective finite code, one occupant per numerical token, the terminal carrier injection, or the endpoint estimate. Generated path population is not an assumption either: it is the theorem `incidencePath_nonempty` already available from F.

8.5. Formal claim discipline

The strongest accurate statement about the present companion is:

> Lean now constructs the complete source-by-source residual-blocker incidence relation and its literal terminal carriers without AASC. It then verifies the choice-independent injective seed-capacity consequence of the explicit relation-level kernel bridge. The manuscript supplies the sunflower-specific proof of that bridge; its direct corpus-to-bridge constructor is the remaining Lean internalization boundary.

This is stronger than the former abstract inheritance interface and narrower than a claim of fully unparameterized machine closure. Both boundaries are visible in the theorem types.

9. Adversarial audit

This section answers the continuation-generation objection at the level of types and dependency order.

9.1. The global residual tower is not source-by-source

Attack. Constructing $\mathcal{F}^{(j+1)}$ does not tell one source $x_j \in B^{(j)}$ how to continue.

Resolution. Accepted and repaired. The source successor set is now defined by the literal formula

$$\mathsf{Next}_j[x_j] = R^{(j)}_{x_j} \cap B^{(j+1)}.$$

It is nonempty because $R^{(j)}_{x_j}$ is an edge of $\mathcal{F}^{(j+1)}$ and the next blocker hits every edge. Theorem 3.18 iterates this finite relation for each source.

9.2. The raw rank step was described rather than defined

Attack. The former symbol δ_j concealed the missing choice of a next blocker coordinate.

Resolution. The symbol and the asserted function have been removed. Definition 3.15 gives the one-step relation, and Definition 3.17 gives the dependent state type. Branching is represented by all tuples satisfying the incidence relation.

9.3. A residual edge is not a blocker point

Attack. The proof may still identify $R^{(j)}_x$ with a point of $B^{(j+1)}$.

Resolution. It does not. The residual is an edge; a successor is a point z satisfying $z \in R^{(j)}_x \cap B^{(j+1)}$. Both types appear separately in the manuscript and in Lean's `GeneratedIncidenceTower.NextCoordinate`.

9.4. Exclusion has been converted into generation

Attack. Eliminating six bad dispositions cannot create the seventh.

Resolution. Correct. Generation is completed first by Theorem 3.18. The disposition theorem classifies members of a pre-existing nonempty finite path set. No-standing-sink is used only to show that the standing subrelation of that generated candidate relation has a nonempty left fibre.

9.5. No-standing-sink need not refer to residualization

Attack. An abstract future continuation need not be one residual rank step.

Resolution. Standing reachability is defined as a subrelation of the specific combinatorial relation Reach_U. A proof of $\mathsf{StReach}_U(x, z)$ contains a generated incidence path witnessing $\mathsf{Reach}_U(x, z)$. The kernel cannot certify a carrier outside that relation.

9.6. Provenance creates a terminal multiplicity dilemma

Attack. If every private lineage remains standing, arbitrarily many lineages may sit over one seed carrier; if lineage becomes skin, injectivity is lost.

Resolution. The proof no longer transports an abstract provenance object and later assigns it to a seed carrier. Every path already ends at a literal seed carrier. The kernel proves directly that a source has one standing terminal carrier and that two same-cell sources cannot share it. Private-edge provenance is used only at the final backward implication: equal complete endpoint type gives source skin-equivalence, and the private edge E_x turns that skin-equivalence into $x = y$.

9.7. The seed blocker was declared primitive-complete

Attack. A constructor atlas cannot prove that every terminal form is one of its proposed carriers.

Resolution. The constructor atlas is no longer used to generate a carrier map. Membership in $B_{\min}(G)$ is part of every path's final coordinate by construction. Theorem 5.11 is now the generatedness theorem $\mathsf{Reach}_U(x, c_U(x))$, not an assertion that an abstract object descends through a convenient blocker.

9.8. The terminal carrier map is not well defined

Attack. A residual may meet several blocker coordinates, so carrierOf has no unique value.

Resolution. The combinatorial relation is intentionally multivalued. Theorem 5.9 proves existence of a standing value, and Theorem 5.10 proves uniqueness of that standing value. Only then is

$$c_U(x) = \iota z\, \mathsf{StReach}_U(x, z)$$

defined. Its value is independent of every representative path.

9.9. Terminal carrier injectivity is assumed

Attack. The 4094 capacity is hidden in a one-occupant-per-carrier premise.

Resolution. Injectivity is proved in Theorem 5.13. A shared carrier inside one fixed cell gives equal complete endpoint type by Theorem 5.12. Same complete type is skin; source skin is literal equality by Proposition 3.4. No numerical code or fibre bound occurs in this proof.

9.10. The profile and role fields are decorative

Attack. If the same collision proof worked on an arbitrary subset, the factor $2^6 \cdot 4$ would do no work.

Resolution. The collision theorem is stated only within one fixed support/profile/role cell. Equality of those fields is used to pass from a shared literal carrier to equality of complete endpoint type. Across distinct profiles or roles, that implication is not asserted. The 256 cells are therefore exactly the domains on which carrier injectivity is proved.

9.11. The number 4094 is kernel-generated

Attack. The finite capacity is an AASC role count.

Resolution. False. It is the ordinary blocker estimate $2 \cdot 2047 = 4094$ and the family-local embedding of Theorem 4.1. AASC proves only that a fixed comparison cell injects into that already finite blocker.

9.12. A universal semantic registry has been assumed

Attack. The same numeral in different families is treated as one standing role.

Resolution. Each terminal family has its own injection into $\mathrm{Fin}(4094)$. Different enumerations differ by a finite permutation. The code is downstream readout, not a global identity registry.

9.13. The finite code is circular

Attack. Code injectivity was used to prove carrier injectivity.

Resolution. The order is the reverse:

generated paths → unique standing carrier → carrier collision exclusion → seed code → counting.

No code occurs in Sections 3 or 5.

9.14. The Lean build proves more than it contains

Attack. Compilation is presented as an unparameterized machine-checked proof of the endpoint.

Resolution. It is not. Lean unconditionally checks the complete generated path relation. The four sunflower-specific governance conclusions remain fields of

`KernelFaithfulStandingPathBridge`.

Lean checks their choice-independent injective and numerical consequences. Section 8 states this boundary explicitly.

9.15. Faithful counterexample worksheet

A counterexample to the crossing must now exhibit at least one of the following concrete failures:

1. a residual edge missed by the next minimal blocker;
2. a source whose generated path set is empty;
3. a live source for which every generated terminal reach is standing-null without target change or endpoint failure;
4. one source with two distinct standing terminal carriers in the same terminal package; or
5. two distinct sources in one support/profile/role cell with one shared standing carrier and no extra tensor, target, or authorizer distinction.

The first two attack ordinary finite combinatorics. The last three attack the named kernel theorems specialized to this endpoint. They do not attack a hidden continuation map or seed atlas.

9.16. Audit conclusion

The repaired crossing is the explicit conjunction

$$\boxed{\text{left-total generated blocker-incidence paths} + \text{ kernel-faithful terminal collision exclusion.}}$$

The first term is constructive finite combinatorics. The second term is AASC exhaustion and impossibility over the candidates produced by the first. Their composition, followed by the independent seed embedding, yields the finite capacity used in the endpoint recurrence.

A. Proof ledgers and reproducibility surfaces

A.1. Symbol ledger

Symbol	Meaning
$B^{(j)}$	Cardinality-minimal literal blocker of $\mathcal{F}^{(j)}$
$E_x^{(j)}$	Private edge of blocker coordinate x at stage j
$R_x^{(j)}$	Residual $E_x^{(j)} \setminus \{x\}$
$\mathsf{Next}_j(x, z)$	$z \in R_x^{(j)} \cap B^{(j+1)}$
$\mathsf{Path}_j(x)$	All generated incidence paths of length j from source x
$\mathrm{car}(P)$	Literal final blocker coordinate of path P
$\mathsf{Reach}_U(x, z)$	Some generated terminal path from x ends at z
$\mathsf{StReach}_U(x, z)$	A generated reach that remains standing-bearing in the fixed endpoint act
$c_U(x)$	Unique standing terminal carrier of source x
$G = \mathcal{F}^{(m)}$	Controlled terminal seed family
$\mathrm{seedCode}_G$	Family-local embedding of $B_{\min}(G)$ into $\mathrm{Fin}(4094)$
slot_U	Composite readout $\mathrm{seedCode}_G \circ c_U$
Q	$2^6 \cdot 4 \cdot 4094 = 1{,}048{,}064$
B	$2^3 Q = 8{,}384{,}512$

A.2. Theorem ladder

1. Core–petal equivalence.
2. Maximal-matching raw blocker.
3. Cardinality-minimal blocker and private edges.
4. Residual-fibre bound and residual-plus-slot injectivity.
5. Residual sunflower lifting and strict rank descent.
6. Nonempty successor intersection with the next blocker.
7. Source-by-source population of all dependent incidence paths.
8. Literal terminal-carrier projection and generated reachability.
9. Kernel activation only after relation population.
10. Standing-subrelation population.

11. Single-valuedness of one source's standing terminal carrier.
12. Same-carrier complete-type determination inside one fixed cell.
13. Private-witness source finality and carrier-map injectivity.
14. Combinatorial terminal-blocker embedding into Fin(4094).
15. Cellwise seed-capacity transport.
16. Same-support finite-code injectivity and fibre bound.
17. Effective blocker and one-point-link recurrence.

A.3. Generation ledger

Generated object	Construction	Nonemptiness reason
Private edge	minimal blocker coordinate x	deleting x would otherwise preserve hitting
Residual edge	$E_x \setminus \{x\}$	private edge exists
Next coordinates	$R_x^{(j)} \cap B^{(j+1)}$	next blocker hits every residual edge
Path set	recursive tuple extension	every next-coordinate set is nonempty
Terminal reach	final coordinate of a path	every source has a path
Seed code	finite embedding of terminal blocker	terminal blocker has at most 4094 points

No row in this table invokes AASC.

A.4. Governance obligation ledger

Obligation	Exact statement	Proof mode
Generatedness	$\mathsf{StReach}_U \subseteq \mathsf{Reach}_U$	definitional typing
Standing population	$\forall x \in U\, \exists z\, \mathsf{StReach}_U(x, z)$	no-standing-sink on populated locus
Choice independence	one source cannot have two standing carriers	split/target/authorizer exhaustion
Complete-type determination	same cell plus same carrier fixes endpoint type	fixed-domain role exhaustion
Quotient finality	same complete type gives source skin	skin has no standing authority
Literal source equality	source skin implies $x = y$	private-edge incidence

These are eliminative statements about generated candidates. None supplies a new path or finite numeral.

A.5. Numerical ledger

$$\begin{aligned} N &= 2047, \\ (3-1)N &= 4094, \\ 2^6 \cdot 4 \cdot 4094 &= 1{,}048{,}064, \\ 2^3 \cdot 1{,}048{,}064 &= 8{,}384{,}512. \end{aligned}$$

The first two values are seed combinatorics. The factors 2^6 and 4 count the comparison cells required by same-carrier complete-type determination. The factor 2^3 counts the conservative ambient support alphabet.

A.6. Circularity audit

None of the following is used to construct Next_j, Path_m, or Reach_U:

- standing or admissibility;
- the support-fibre inequality;
- a finite global code;
- a bounded-load object;
- terminal-carrier injectivity; or
- the endpoint theorem.

None of the following is used to prove terminal-carrier injectivity:

- the seed enumeration;
- the number 4094;
- the support-fibre bound;
- the effective blocker bound; or
- the endpoint recurrence.

The dependency order is

$$\text{generated relation} \to \text{standing subrelation} \to \text{unique injective carrier} \\ \to \text{numerical readout} \to \text{counting.}$$

A.7. Claim-status and nonclaim register

- The theorem is not a new asymptotic sunflower bound.
- A residual edge is not identified with a lower blocker point.
- No Hall matching or canonical successor is used.
- The number 4094 is not generated by AASC.
- Numerical seed labels are not universal semantic roles.
- The former abstract inherited-form and seed-atlas maps are not used.
- Lean has internalized the complete positive path generation.

- Lean has not yet internalized the direct corpus-to-governance-bridge constructor.
- A failure of standing population, single-valuedness, or same-carrier complete-type exhaustion falsifies the paper-level crossing.

A.8. Reproducibility checklist

The publication package contains:

1. the complete LaTeX source and bibliography;
2. the final PDF;
3. the supporting corpus papers used by the theorem audit;
4. citation, theorem-use, and build audits; and
5. a SHA-256 integrity manifest.

The build command is:

```
./build.sh
```

A publication build is accepted only if it has no undefined references or citations, no overfull boxes, embedded fonts, successful PDF preflight, and a clean rebuild from the delivered archive.

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